OBJECTS OF A RESOLVING F-EXACT SUBCATEGORY ARE RIGHT ACYCLIC

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 5640-5677, following the statement at 5624-5638.

Editorial note. The 0-in-I normalization is handled in an explicit footnote (5646-5649), recorded.

Allowed standard prerequisites (editorial).

- Right resolutions by the subcategory. lemma-subcategory-right-resolution, cited at proof line 5646; short source lemma (statement-level input, proof at 5620-5622 minimal). Supplies the quasi-isomorphism K -> I with I^n in the subcategory, giving cofinality of I-resolutions.

Source reference. Original lemma 05T6: sources/stacks-derived/pinned-original-source.tex, source lines 5604–5604. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 05T4: sources/stacks-derived/pinned-original-source.tex, source lines 5410–5410. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *In Situation 05T4. Let $\mathcal{I} \subset \mathrm{Ob}(\mathcal{A})$ be a subset with the following properties:*

1. *every object of $\mathcal{A}$ is a subobject of an element of $\mathcal{I}$,*
2. *for any short exact sequence $0 \to P \to Q \to R \to 0$ of $\mathcal{A}$ with $P, Q \in \mathcal{I}$, then $R \in \mathcal{I}$, and $0 \to F(P) \to F(Q) \to F(R) \to 0$ is exact.*

Then every object of $\mathcal{I}$ is acyclic for RF.

Proof. Pick $A \in \mathcal{I}$. Let $A[0] \to K^\bullet$ be a quasi-isomorphism with $K^\bullet$ bounded below. Then we can find a quasi-isomorphism $K^\bullet \to I^\bullet$ with $I^\bullet$ bounded below and each $I^n \in \mathcal{I}$, see Lemma 05T6[1]. Hence we see that these resolutions are cofinal in the category $A[0]/\mathrm{Qis}^+(\mathcal{A})$. To finish the proof it therefore suffices to show that for any quasi-isomorphism $A[0] \to I^\bullet$ with $I^\bullet$ bounded below and $I^n \in \mathcal{I}$ we have $F(A)[0] \to F(I^\bullet)$ is a quasi-isomorphism. To see this suppose that $I^n = 0$ for $n < n_0$. Of course we may assume that $n_0 < 0$. Starting with $n = n_0$ we prove

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[1]By (1) we see that $\mathcal{I}$ is nonempty. Pick P in $\mathcal{I}$. Then the short exact sequence $0 \to P \to P \to 0 \to 0$ and assumption (2) shows that 0 is in $\mathcal{I}$. Thus the lemma applies.

inductively that $\mathrm{Im}(d^{n-1}) = \mathrm{Ker}(d^n)$ and $\mathrm{Im}(d^{-1})$ are elements of $\mathcal{I}$ using property (2) and the exact sequences

$$0 \to \mathrm{Ker}(d^n) \to I^n \to \mathrm{Im}(d^n) \to 0.$$

Moreover, property (2) also guarantees that the complex

$$0 \to F(I^{n_0}) \to F(I^{n_0+1}) \to \ldots \to F(I^{-1}) \to F(\mathrm{Im}(d^{-1})) \to 0$$

is exact. The exact sequence $0 \to \mathrm{Im}(d^{-1}) \to I^0 \to I^0/\mathrm{Im}(d^{-1}) \to 0$ implies that $I^0/\mathrm{Im}(d^{-1})$ is an element of $\mathcal{I}$. The exact sequence $0 \to A \to I^0/\mathrm{Im}(d^{-1}) \to \mathrm{Im}(d^0) \to 0$ then implies that $\mathrm{Im}(d^0) = \mathrm{Ker}(d^1)$ is an elements of $\mathcal{I}$ and from then on one continues as before to show that $\mathrm{Im}(d^{n-1}) = \mathrm{Ker}(d^n)$ is an element of $\mathcal{I}$ for all $n > 0$. Applying F to each of the short exact sequences mentioned above and using (2) we observe that $F(A)[0] \to F(I^\bullet)$ is an isomorphism as desired. $\square$